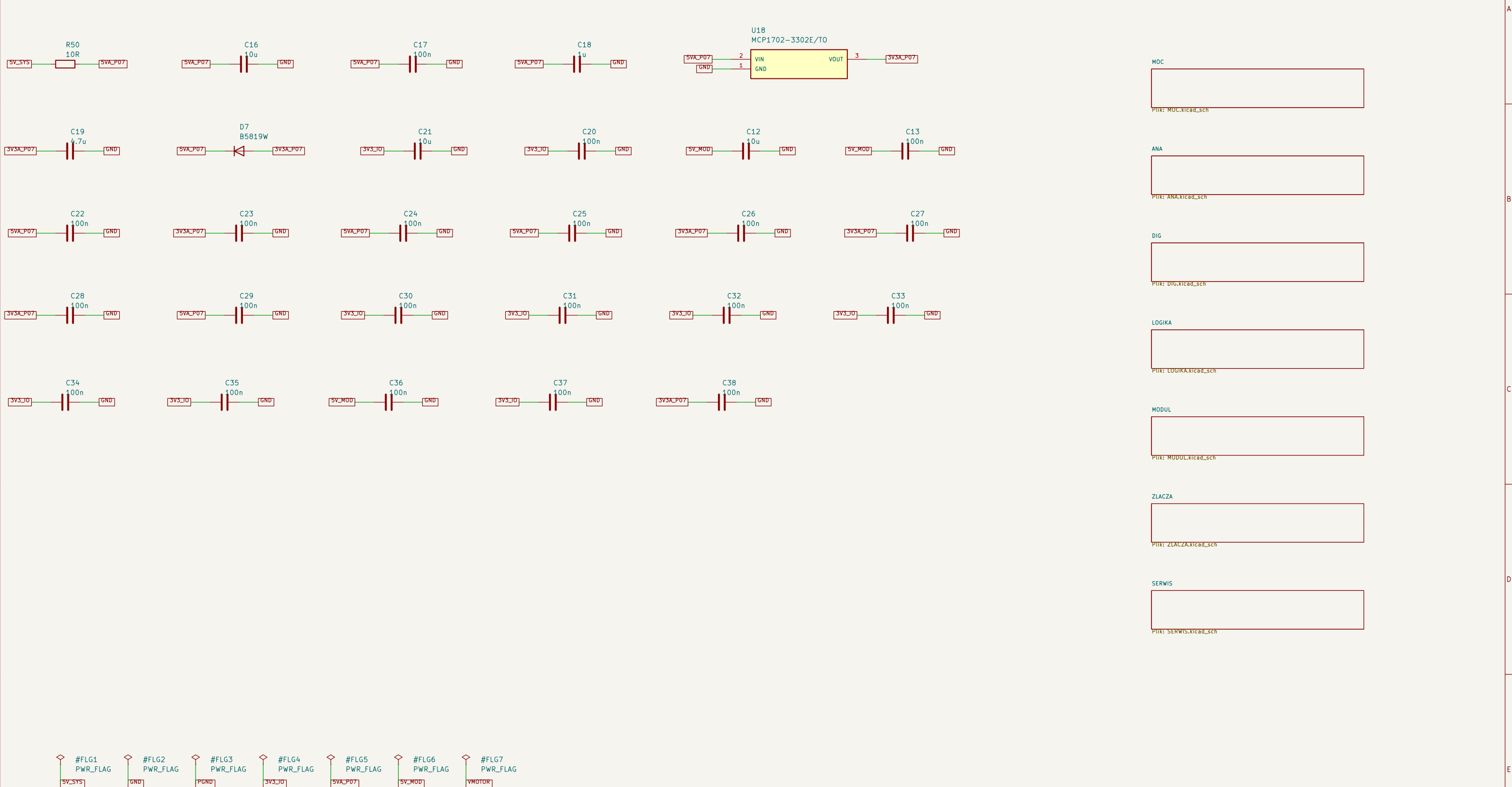
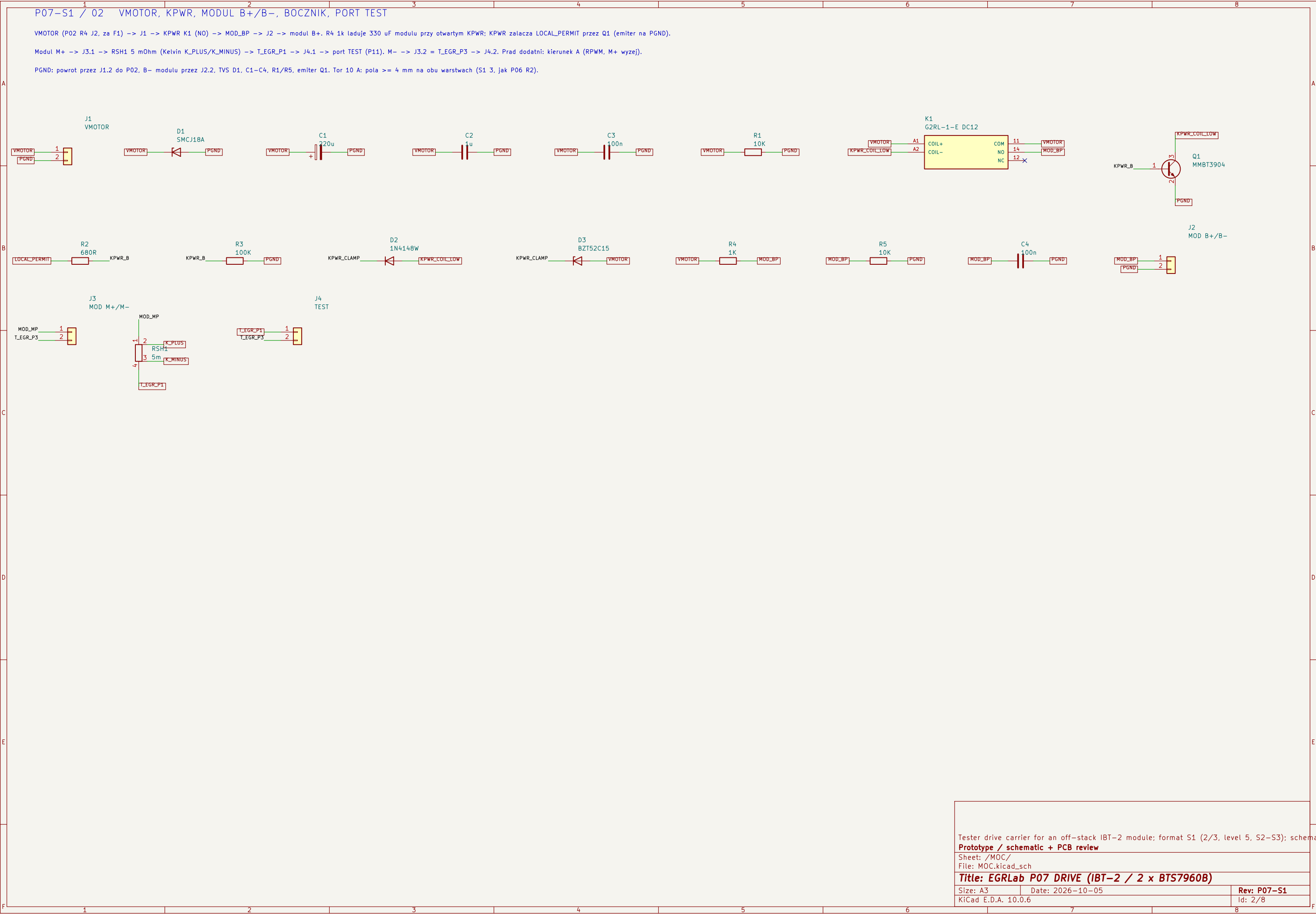


P07-S1 / 01 ZASILANIE LOKALNE I ARKUSZE

J_BP2: 5V_SYS (piny 10/12) i 3V3_I0 (pin 14) z P02 R4 przez P12. Logika (U10–U15, U17) na 3V3_I0 – ta sama szyna co P04 R3. Analog na lokalnych 5VA_P07 (R50 10R) i 3V3A_P07 (U18).

5V_MOD = 5V_SYS za PTC F1: VCC modulu IBT-2 (74HC244) i bufora U16. GND (logika, analog) i PGND (prąd silnika) NIE są połączone na P07: wspólny punkt mas = P02 R4 J2 (VMOTOR).



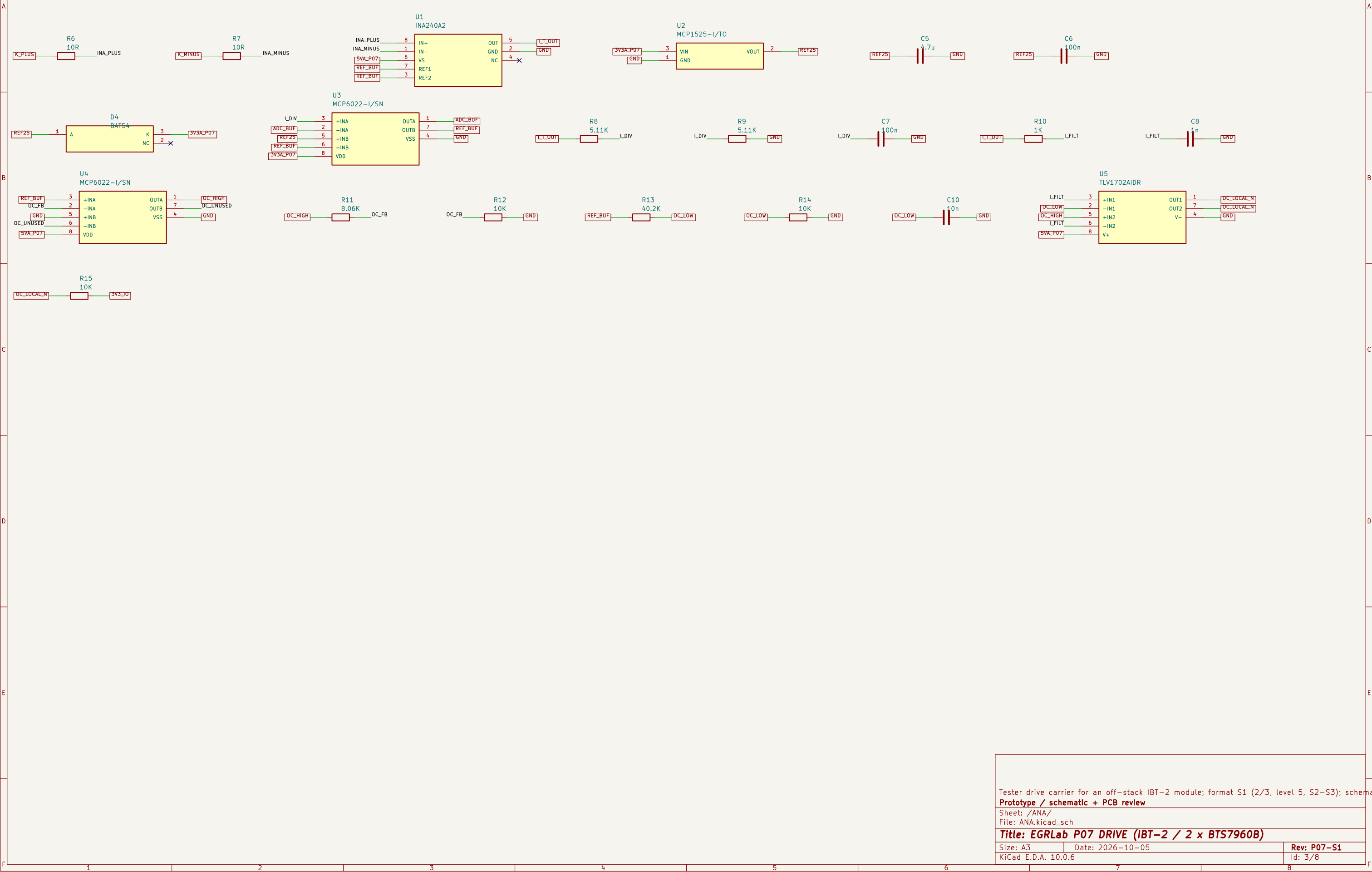


Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only.		
Prototype / schematic + PCB review		
Sheet: /MOC/		
File: MOC.kicad_sch		
Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)		
Size: A3	Date: 2026-10-05	Rev: P07-S1
KiCad E.D.A. 10.0.6		Id: 2/8

P07-S1 / 03 INA240, ODNIESIENIE, ITEST, OKNO OC

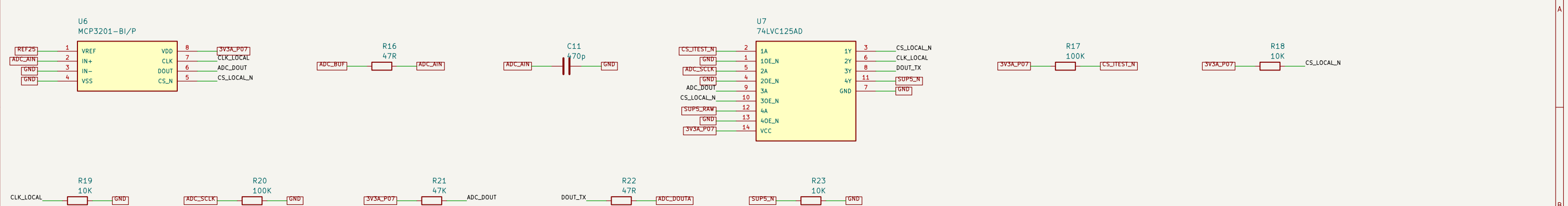
LT_OUT = REF_BUF + 50 x 5 mOhm x I = 2,5 V + 0,25 V/A. ITEST: R8/R9 1:2 + C7 100n (tau 0,26 ms) -> U3A -> MCP3201: 1,25 V + 0,125 V/A, +-10 A w 0..2,5 V.

Okno OC: LFILT (R10 1k / C8 1n) vs OC_HIGH = REF_BUF x 1,806 (4,515 V, +8,06 A) i OC_LOW = REF_BUF x 0,1992 (0,498 V, -8,01 A), U5 open collector -> OC_LOCAL_N (R15 10k do 3V3_IO).



P07-S1 / 04 MCP3201 ITEST I BUFOR SPI

SPI mode 0, 16 zegarów, raw = (word >> 1) & 0x0FFF (jak P06 R2). DOUT_TX trojstanowy (OE = CS_LOCAL_N) + R22 47R na wspólnej ADC_DOUTA. U7 gate 4: SUP5_RAW (5 V) -> SUP5_N (3,3 V).

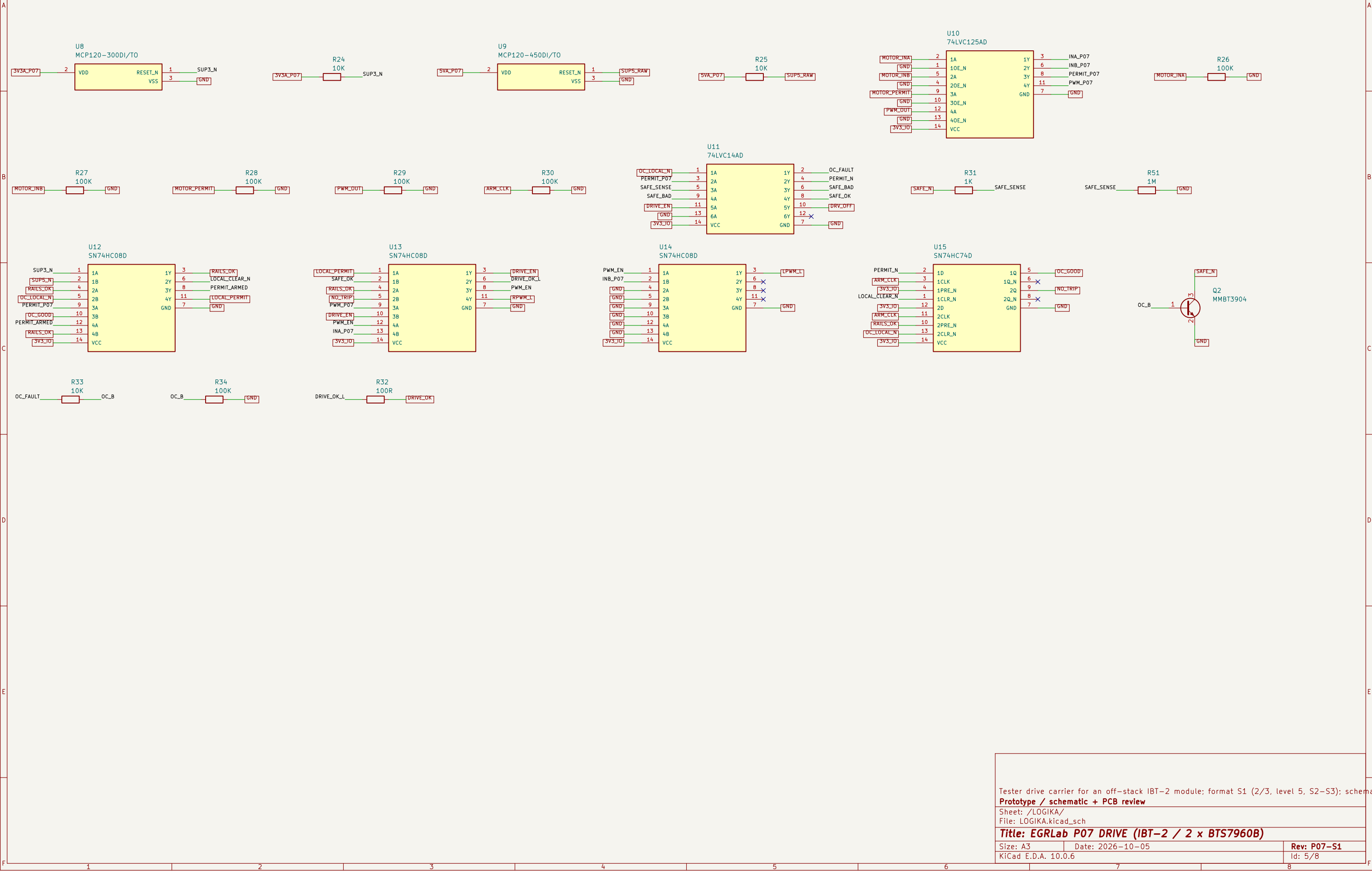


P07-S1 / 05 NADZORCY, ODBIORNIKI, BRAMKI, ZATRZASK OC, SAFE_N

RAILS_OK = SUP3_N & SUP5_N. LOCAL_CLEAR_N = RAILS_OK & OC_LOCAL_N. OC_GOOD: kasowany przez OC/zasilanie, ustawiany TYLKO zboczem ARM_CLK przy MOTOR_PERMIT = L (D = PERMIT_N).

LOCAL_PERMIT = MOTOR_PERMIT & OC_GOOD & RAILS_OK (-> KPWR). DRIVE_EN = LOCAL_PERMIT & SAFE_OK. RPWM = PWM_OUT & DRIVE_EN & MOTOR_INA; LPWM = PWM_OUT & DRIVE_EN & MOTOR_INB.

NO_TRIP (FF2): PRE = RAILS_OK (start = brak zadzialania), CLR = OC_LOCAL_N, D = 1 przy ARM_CLK. DRIVE_OK = RAILS_OK & NO_TRIP. Q2 sciaga SAFE_N podczas OC.

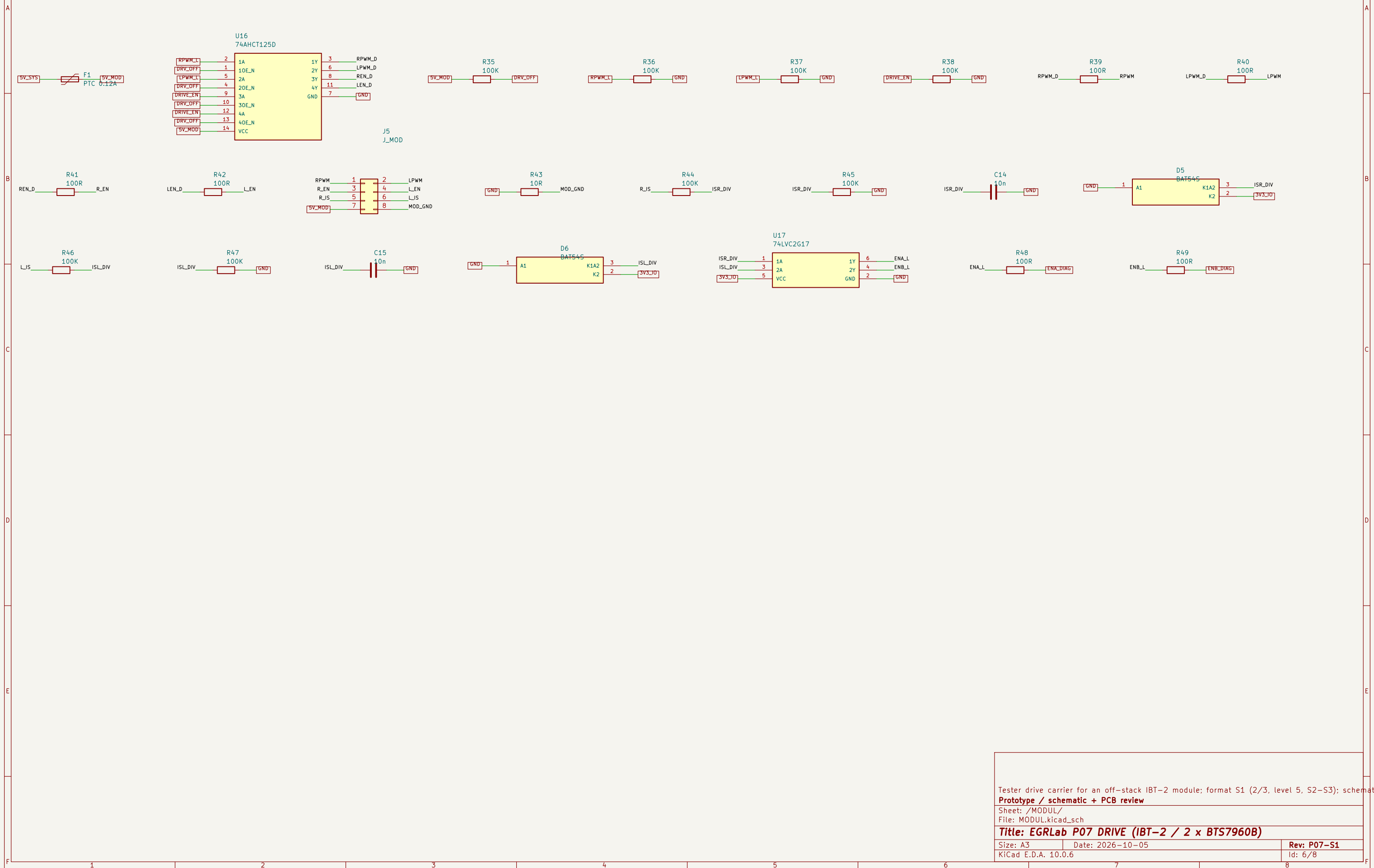


P07-S1 / 06 BUFOR 3,3 -> 5 V, WIAZKA MODULU, DIAGNOSTYKA IS

U16 74AHCT125 (5V_MOD): 3,3 -> 5 V dla 74HC244 modulu (VIH 3,5 V przy 5 V). OE_N = DRV_OFF; wyjścia Z bez DRIVE_EN; modul ma 30k do GND na wejściach -> L, R_EN = L_EN = DRIVE_EN.

IS: modul 10k do GND (ok. 1,2 V/A). R44/R45 1:2 + C 10n, clamp BAT54S do 3V3_I0, U17 Schmitt -> ENA_DIAG / ENB_DIAG (H = prąd galezi >-2-3 A albo poziom błędu IS). Tylko diagnostyka.

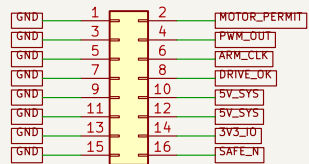
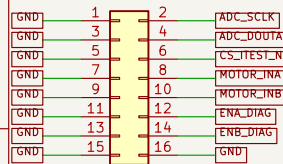
MOD_GND przez R43 10R: działa przy GND modulu połączonym z B- i przy rozdzielonym (pomiar 5.10: 508 mV w trybie diody).



P07-S1 / 07 J_BP1 / J_BP2 DO P12 (KRAWEDZ A)

J_BP1 (S2, x płytki 26,5 / stosu 80,0): ADC_SCLK 2 / ADC_DOUTA 4 jak J_BP2 P03 R6, P05 R3 i J_BP P06 R2; CS_ITEST_N, MOTOR_INA/INB, ENA/ENB_DIAG do P03 R6.

J_BP2 (S3, 4, 5) 80,0 / stosu 133,5); MOTOR_PERMIT 2, PWM_OUT2 4, ARM_CLK 6, DRIVE_OK 8, SAFE_N 16 na tych samych pinach co J_BP2 P04 R3; 5V_SYS 10/12, 3V3_IO 14. Nieparzyste GND.



Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only. Prototype / schematic + PCB review	
Sheet: /ZLACZA/ File: ZLACZA.kicad_sch	

Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)

Size: A3	Date: 2026-10-05	Rev: P07-S1
KiCad E.D.A. 10.0.6		Id: 7/8

P07-S1 / 08 LISTWY SERWISOWE J_SV1 / J_SV2 (KRAWEDZ B)

J_SV1 (S2, x 10..43); wezly analogowe 10k (piny 2-7), szyny pakietu 4,7k (9-12), GND 1/8/13. J_SV2 (S3, x 63.5..96.5); szyny 1k (2-6), logika 1k (8-12), GND 1/7/13.

Rezystor przy wezle (S1 6). Szyna tylko obok GND, innej szyny albo linii logicznej; wezly analogowe tylko obok siebie i GND.

